

An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

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Thapar Institute of Engineering & Technology · August 22, 2026

1. Executive summary

Floods displace millions of people in India every year, and the slow part of a rescue is not reaching survivors — it is *finding* them.

We are building three one-metre, 6.4 kg multirotor aircraft that search flooded ground as a coordinated group under **one operator**, recognise people from the air, report their position to about a metre, and drop a relief kit to each. It requires **no mobile network and no internet**, which matters because those are among the first things a flood removes.

The sizing, error budgets and flight software are complete and reproducible, and coordination has been demonstrated across three simulated autopilots. No aircraft has flown yet, so the numbers here are calculated; the plan below is ordered to replace them with measurements, hardest question first.

We are asking for **INR 7,43,004 across 32 steps**, none above INR 30,000, each released only on the result of the one before. The first is INR 26,196.

2. The problem

After a flood the binding constraint is time-to-locate. Boat teams search slowly, at high risk, and cannot see onto rooftops. Helicopters are scarce, costly per hour, and committed to evacuation. Nothing covers a large area quickly *and* reports exactly where people are.

Small drones look like the answer and mostly are not: they need one pilot each, depend on a network the disaster has removed, and find people without helping them.

The harder problem is seeing a person at all. Someone in floodwater shows head and shoulders only, about 40 cm across. Detectors take a fixed, small input, so the conventional approach shrinks the image to fit — and the person shrinks with it, to a size where published results report only 29–61 % accuracy. Choosing a smaller, faster model makes this *worse*, because smaller models take smaller inputs. Two further constraints are specific to floods: no existing dataset labels *people* in flood imagery, and thermal cameras fail in water, where skin cools to water temperature.

3. Our solution

The system. The operator draws a search area of any shape; the software divides it into three balanced pieces and plans a path over each — verified on thirteen awkward boundary shapes with no path leaving the region and under 0.02 % workload imbalance. The aircraft then fly themselves, and our test harness proves structurally that no command is sent after the mission begins. Detection and geolocation run *on board*, so nothing depends on a link. Each aircraft carries four kits and can descend, stop over a survivor and release one. Link loss, low battery and operator abort all converge on the same return-and-land behaviour.

Seeing the survivor. We do not shrink the image. The full-resolution frame is cut into 48 overlapping tiles, each examined at native resolution, and we fly at 40 m rather than 60 m. Together these put the person at **39 pixels** instead of 13 — nine times the area, and out of the size band where detectors perform worst. Only both changes together achieve it; either alone leaves the target in the difficult band.

It costs four times the computing, and it fits: 48 tiles at 2 Hz is 96 inferences per second against the 130–160 measured for our accelerator. The extra transects cost seconds, and the mission is in fact slightly shorter than at 60 m because the climb is lower.

One item is open. Confirming a survivor across twelve frames needs about 3 Hz, which does not fit. A two-stage approach would buy it back — a cheap model skipping the roughly 87 % of tiles that are empty water — and we will test it on free public data before adopting it, because a first-pass filter that discards a tile containing a person fails silently.

4. Deliverables

1. Three aircraft flying as a coordinated group under one operator.
2. A ground station that displays everything and is deliberately *unable* to fly the aircraft, so the autonomy cannot be undermined in use.
3. Autonomy software — area division, path planning, allocation, failsafes — with an automated test suite.
4. A *measured* detection figure, replacing the calculated one.
5. A measured answer on whether the two-stage filter can raise the capture rate enough to confirm survivors across twelve frames.
6. A publication: image tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
7. Code and results published either way.

5. Benefits

Operational. Three aircraft sweep ten hectares in under eight minutes and hand a coordinator a list of coordinates rather than a video feed somebody must watch — with no network, which is the condition that actually obtains.

Economic. The whole system costs less than one hour of helicopter search time, travels by road, and is run by two people.

Departmental. A calibrated thrust stand, a certified scale, a ground station, a tested autonomy codebase and trained students all outlive the project.

Strategic. Indian suppliers wherever they meet the specification — for sovereignty, and because domestic lead times are half those of imports, which buys flight-test days.

Transferable. Nothing is flood-specific except the training data. The same system suits earthquake rubble survey, wildfire mapping and avalanche search.

6. Resources needed

Every part below is a specific model at a current Indian retail price, chosen against a number the design produces. Where a cheaper part exists and was rejected, the reason is given.

Measurement equipment — bought first, because it gates everything

Item	Model / specification	INR	Why this, and why this specification
Thrust stand	20 kg load cell, HX711 24-bit amplifier, inline power meter, frame fabricated in-house	8,000	Our motors are sold without a thrust curve, so the single most important number in the design — whether four of them lift 6.36 kg — is currently unknown. The load cell resolves to a few grams against a 3.18 kgf requirement, and measuring current at the same time gives thrust <i>per watt</i> , which is what sets endurance. Commercial stands start near INR 45,000; the sensing parts cost under INR 1,000 and the frame is a machine-shop job.
Calibrated scale	Bench scale, 30 kg capacity, 1 g resolution, with calibration certificate	12,000	Maximum take-off mass is a hard limit, not a target. 1 g resolution on a 6 360 g aircraft is 0.02 %, so build-up can be tracked part by part rather than discovered at the end. The certificate is the point: an uncertified scale gives a number we cannot defend.

Per aircraft — avionics and sensing INR 85,200

Item	Model	INR	Why this, and why this specification
Flight controller	Holybro Pixhawk 6C Mini	22,600	Two independent inertial sensors on silicone isolation, so a single sensor failure or a resonance does not end the flight. Runs ArduPilot natively, which is what our simulation testing is built on. Cheaper F4-class boards exist at around INR 6,000 but lack the memory headroom for the logging and failsafe configuration we rely on.
GNSS receiver	RTK-capable, RTCM3 corrections, ≤ 3 cm CEP	18,000	Position accuracy is the single largest error term in the system. With ordinary GNSS a reported survivor position is good to about 3.9 m; with RTK it is 0.75 m. Nothing else in the budget improves the number comparably, which is why this line survives every cost reduction.
Onboard computer	Raspberry Pi 5, 8 GB	8,000	The host everything runs on: Linux, the coverage planner, message routing, the mesh link and the delivery logic. It is also the only board in this class that exposes PCIe, which the accelerator physically requires — the previous generation cannot be used at all.
AI accelerator	Raspberry Pi AI HAT+ 26 TOPS (Hailo-8)	11,000	Runs the detector on board, so detection never depends on a radio link. Measured at 130–160 frames per second for our model and input size; we need 96. The 13 TOPS version is about INR 4,000 cheaper but manages 60–80, which does not cover 96.
Camera + lens	Arducam IMX477, 6 mm CS lens, fixed focus	11,100	1.55 μm pixels give 1.03 cm per pixel at 40 m, which is what puts a person in water at 39 pixels. Fixed focus is deliberate: the hyperfocal distance is 4.15 m, so a lens set once is sharp from 2 m to infinity and we never fly below 30 m. A motorised-focus module costs more, adds a moving part to a vibrating airframe, and can drift without any indication from the air.
Radio + storage	ExpressLRS 2.4 GHz receiver (firmware 3.5+), storage, cooling, mounts	8,500	In its native mode this one link carries manual control <i>and</i> telemetry on a single autopilot port. That is why no second radio is needed — a saving of about INR 19,000 per aircraft — and the firmware version matters, because the older mode consumes the link and would leave the aircraft with telemetry and no control.
Mesh radio	5.8 GHz node with antennas	6,000	Carries three simultaneous video feeds to the ground station. Total offered load is 2.5 Mbps against a 20 Mbps channel; the constraint is link margin at range, not bandwidth.

Per aircraft — airframe, power and drive INR 74,100

Item	Model	INR	Why this, and why this specification
Motors	Tarot TL96020, 5008 frame, 340 KV, 4 off	18,000	Must produce 3.18 kgf each at full throttle so the aircraft has twice its weight in thrust, with hover at half throttle where propellers are efficient and there is authority left for gusts. 340 KV is the winding that suits an 18 in propeller on this battery voltage.
Airframe	25 × 23 mm carbon tube, machined clamps, plate; fabricated in-house	13,000	The tube is sized so bending stress at full thrust is about 4 % of what the material takes — bending is not what breaks these aircraft; clamp design and vibration fatigue are, which is why the joints are machined rather than printed.
Battery cells	Molicel P45B 21700, 4500 mAh, 45 A, 18 off	12,600	Eighteen cells in a six-series, three-parallel pack: 292 Wh, enough for the mission plus a complete second search plus four minutes of loiter. Peak draw is 38.3 A per cell, so a 40 A cell would run at 96 % of rating with no margin. Capacity and internal resistance must be matched, because one weak cell sets the behaviour of the whole pack.
Pack assembly	6S 60 A battery management, power distribution, regulator	8,500	The cells arrive loose. This is the welding, fusing, balancing and monitoring that turns them into a pack that can be charged and discharged safely.
Speed controllers	50–60 A continuous, 6S, 4 off	7,200	Peak demand is 29 A per motor, so a 50–60 A controller runs at about half rating and stays cool. Undersizing here is a common cause of mid-flight failure.
Propellers	Tarot 1855 carbon, CW/CCW, 4 off	6,500	18 in carbon matched to the motor and battery voltage. Verified at INR 3,250 per counter-rotating pair. These are the most frequently destroyed item in any flight-test campaign and should be treated as semi-consumable.
Payload release	Four metal-gear servos with mechanical detents	4,500	Metal detents rather than servo torque alone, so a power interruption cannot release a kit. Four stations, released one at a time.
Wiring	10 AWG silicone main leads, XT90-S anti-spark, JST-GH signal	2,400	Gauge is set by the 115 A peak; thinner wire would drop voltage and heat under load. The anti-spark connector prevents the arc that pits contacts when a 292 Wh pack is plugged in.
Propeller adapters	6 mm self-tightening hubs, 4 off	1,400	Self-tightening so rotation seats the nut rather than loosening it. A propeller released at 5,000 rpm destroys the aircraft.

Flight-test support INR 36,500

Item	Model	INR	Why this, and why this specification
Battery chargers	ToolkitRC M6D, 500 W, 25 A, dual channel, 2 off	11,672	Our pack is 13.5 Ah; charging it in about an hour needs roughly 340 W. The commonly recommended HOTA D6 Pro (INR 10,499–INR 13,500) delivers 325 W per channel and would be at its limit. The M6D delivers 500 W at INR 5,836 and needs only an external supply, which the department already owns. Two units give four channels, so the whole fleet charges in parallel on a flight day.
Consumables	Fasteners, thread-lock, carbon tape, heat-shrink, printer filament	13,000	Consumed continuously through assembly and repair. Running out stops work, and these are the items nobody remembers to budget.
Safety-pilot radio	RadioMaster Boxer, ExpressLRS	6,955	Manual override, required for every test flight. Full-size hall-effect gimbals rather than a pocket radio, because a pilot taking control of a 6.4 kg aircraft in an emergency needs proper sticks.
Sun hood + monitor	Fabricated hood, second-hand monitor	1,500	Laptop screens are unreadable in Punjab daylight, and an operator who cannot see the display cannot supervise the flight.

Ground segment and statutory INR 52,000

Item	Model	INR	Why this, and why this specification
Field consumables	Cables, mounts, field spares, transport protection	25,000	Position corrections travel over the telemetry link we already have, so no separate base-to-aircraft radio is needed.
Base station receiver	Second GNSS receiver, same class as the aircraft unit	18,000	Centimetre positioning works by comparing the aircraft against a stationary receiver on a surveyed point. Without it the aircraft fall back to ordinary GNSS and reported positions degrade from 0.75 m to 3.9 m.
Registration	Statutory registration, three aircraft	5,000	At 6.36 kg these fall in the Small category of the Drone Rules 2021, so registration is mandatory before any flight.
Survey tripod	Photographic tripod and adapter	4,000	The base receiver needs to be stationary and repeatable, not surveyed to professional tolerance — the corrections are relative.

Release schedule — 32 steps, none above INR 30,000

The order is engineering, not accounting: prove the motor before buying twelve, and fly one complete aircraft before building the other two.

Steps	Content	INR	Released only when
1–2	Thrust stand, scale, and one motor to measure	31,371	On approval
3–10	Aircraft one, in eight purchases	1,97,238	Measured thrust ≥ 3.18 kgf per motor
11–12	Chargers, safety radio, consumables, hood	42,574	Airframe assembled and weighed
13–16	Base station, tripod, field kit, registration	66,992	First flight completed
17–24	Aircraft two, same eight purchases	2,02,413	Aircraft one flew a full mission
25–32	Aircraft three, same eight purchases	2,02,413	Two aircraft flying, separation verified
Total		7,43,004	

No individual step exceeds INR 29,440. Aircraft two and three are the last requests, so by the time either is made a complete aircraft has already flown the mission — what is funded is replication of a working design. (Steps are rounded to the rupee, so the column sums to within a few rupees of the total.)

What this structure costs. Thirty-two approvals is real administrative work, and buying one aircraft at a time means placing the same imported order three times, worth roughly eight to twelve weeks of schedule. We have chosen it anyway; if either becomes binding, merging the per-aircraft steps into one order is the lever.

The detection test needs nothing new — about one week on a departmental GPU and 30 GB of storage. The dataset is public and the software is open source.

7. Future work

Near term. Replace every calculated number with a measured one: detection accuracy, motor thrust, setup time. Test near-infrared sensing — water absorbs infrared and people do not, so a person appears bright against dark water. A camera without its infrared filter costs about INR 3,000 and a public dataset exists to test the idea before buying. Beyond floods, the same system suits earthquake, wildfire and avalanche search.

Longer term: aircraft that must think for themselves. Our central constraint — a capable aircraft deciding for itself inside a very small power budget — is a hard requirement in planetary aviation. NASA’s *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth’s density. *Dragonfly* launches in 2028 for Titan, where the air is four times denser than Earth’s at one seventh the gravity.

What makes those hard is not the flying but that **a radio command takes minutes to arrive**: nobody is piloting, so every choice is made on board. What transfers from this work is what is not about floods — planning a survey on board, running perception to an *energy* budget, and failing safely with no link. One caution worth stating: microcontroller-class processors, the hardware usually proposed for such missions, cannot resolve targets of the size we care about, so a planetary version needs a different operating point. That is a research question in its own right.

8. Conclusion

The design is complete and explicit about which numbers are measured and which are calculated. We are asking for the means to finish it in 32 steps, none above INR 30,000, so nothing is committed before the previous step has produced evidence.

The first step is INR 31,371 and answers the question everything else depends on: can these motors lift this aircraft. The immediate technical need is smaller still — a week of GPU time to test the detection approach on free public data.

We would value your guidance on two decisions: the operating altitude, which the detection test will inform, and whether to adopt the two-stage detector once we have measured it.